

ECE4560 - Introduction to Automation and Robotics (4-3-3)

School of Electrical and Computer Engineering
Georgia Institute of Technology

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<i>Inquiry Room:</i>	TSRB 442	<i>Inquiry Room:</i>	van Leer E265
<i>Inquiry Hours:</i>	Friday 10:00AM - 11:00AM	<i>Inquiry Hours:</i>	Thursday 4:00PM - 6:00PM

Laboratory: VL E265
Lab Hours: Open Lab from 8AM to 8PM (requires Card Access after)

Prerequisite: ECE 3085/3084 or ECE3550

Optional Books: Craig, J.J. Introduction to Robotics: Mechanics and Control, 3rd Ed.
Murray, Li, and Sastry. A Mathematical Introduction to Robotic Manipulation.
Lynch, and Park. Modern Robotics: Mechanics, Planning, and Control.
Spong, Hutchinson, Vidyasagar. Robot Modeling and Control. 3rd Edition.
Additional eBooks to be noted elsewhere.
Purchase not required.

Catalogue Description: Fundamental disciplines of modern robotics: mechanics, control, and computing. Analysis, design, and control of mobile robots and manipulators. Course may contain team projects and hands-on labs.

Scope and Goals: The goal of the course is to provide you with the knowledge to analyze and understand robotic manipulators. Consequently the course will cover elements of theoretical and applied kinematics, which deal with the mathematical analysis and synthesis of mechanical manipulators.

At the termination of the class, you should be able to:

- analyze and mathematically describe a given manipulator,
- understand said manipulator's operational limits,
- create design proposals for a manipulator given its specifications,
- confidently read the kinematics and robotics literature, and possibly
- understand some of the programmatic issues related to serial manipulators.
- potentially pursue and demonstrate mastery in an area of robotics of your choosing.

Topic Outline: The course is broken up into two segments covering the following topics:

1. Mathematics and Modeling.

- Coordinate representation of manipulators
- Homogeneous coordinates and representation of orientation
- Lie groups and Lie algebras, exp and ln
- Body versus spatial reference frames.

2. Kinematics

- Forward kinematics
- Workspace analysis
- Inverse kinematics
- Manipulator velocities and Jacobian
- Task planning

3. Dynamics

- A primer on Euler-Lagrange equations, Lagrangian mechanics
- Position/Torque control

Course Mechanics and Grading: The course meets two times a week, TuTh 5:00-6:15PM in Mason 3133. Being a laboratory course, each student must identify a lab experience. This will be discussed in the following section.

The course grading criteria consist of the following components whose percentage of the total grade calculation is also given,

	Homework	Lab Checkpoints	Final Project	Midterm 1	Midterm 2	Final
Percent of Final Grade	10%	10%	10%	20%	20%	30%

where the contributions are based on total percentage per category (e.g., total percentage across all homeworks).

Lab Projects. The lab project will culminate in a final presentation. The labs will be group-based with each group consisting of 2-3 students. The weekly labs are all intended to be used with the SO-101 manipulator arms, but there are also three legacy platforms that students can choose to use for the final project if desired. These legacy platforms are the Piktul planar manipulator, Turtlebot mobile robot, and a planar Bipedal robot. Each group will be formed and asked to select a lab project track within the first few weeks of the course.

Homeworks. The homeworks are intended to reinforce the topics presented in class. I will typically make an effort to be available for a short period after class for any questions that may arise. You should also feel free to e-mail me if homework statements are unclear or to set up an alternative time to meet, however please be sure to include the string “ECE4560” in the subject heading.

Homework will be assigned approximately every week. These homework assignments will be due on Friday (11:59pm). All material should be submitted virtually on Gradescope via Canvas.

Solutions for the homework will be posted 24 hours after the deadline to allow for late submissions.

All students are automatically given three total days of permitted late extensions. These days can only be used in increments of 24 hours. If you need to use an extension that is longer than 24 hours, you must tell Dr. Tucker to prevent the homework solutions from being posted prematurely. Once your three extension days have been used, homeworks can be submitted up to 24 hours late, without request, for a 20% penalty. However, after solutions have been posted, no further late submissions will be accepted. If you have any questions about the late submission policy, please contact Dr. Tucker.

While collaboration is encouraged regarding the homework material, all work to be turned in is expected to be individually completed. It is presumed that we are all operating under the Georgia Tech Honor Code.

Programming. The course will require programming the learned mathematics into Matlab and/or Python. If you are unfamiliar with either Matlab or Python, there are many sources of documentation and primers online.

The SO-101 manipulator project will be implemented in Python. For those choosing to work with the Piktul manipulator, Matlab will be the only option provided. Likewise, the bipedal robot project will involve Matlab plus some additional libraries that can be compiled to run on Windows, MacOS, and Linux. The Turtlebot Mobile Robot lab thread will involve the Robot Operating System (ROS) and python running on the Linux operating system.

Course Schedule (subject to change):

Course Schedule			
Week	Date	Lecture	Assignment Due
1	August 25	Syllabus + Introduction	
	August 27	1. Planar Kinematics	
2	September 1	2. Coordinate Frames & Rotations	
	September 3	3. Transformations	HW 1
3	September 8	4. Homogeneous Coordinates	
	September 10	5. Velocity of a Rigid Body	HW 2
4	September 15	6. Exponential Representation of Rotations	
	September 17	7. Exponential Representation of Motion	HW 3
5	September 22	8. Velocity of a Screw Motion	
	September 24	9. Review	HW 4
6	September 29	– Midterm 1 Exam –	
	October 1	10. Manipulators & Manipulator Analysis	
7	October 6	– Fall Break –	
	October 8	11. Forward Kinematics	HW 5
8	October 13	12. Inverse Kinematics 1	
	October 15	13. Inverse Kinematics 2	HW 6
9	October 20	14. Manipulator Jacobian 1	
	October 22	15. Manipulator Jacobian 2	HW 7
10	October 27	16. Review	
	October 29	– Midterm 2 Exam –	
11	November 3	17. Jacobians and Singularity Analysis	
	November 5	18. Trajectory Design 1	
12	November 10	19. Trajectory Design 2	
	November 12	20. Trajectory Design 3	HW 8
13	November 17	21. Trajectory Design 4	
	November 19	22. Wrenches and Forces	HW 9
14	November 24	23. Introduction to Control	
	November 26	– November Break –	
15	December 1	Group Projects	
	December 3	Group Projects	HW 10
16	December 8	Final Exam Review	
	December 10	– Final Exam (6-8:50pm) –	